

6.7980: Topics in Multiagent Learning

Lecture 2 – Prof. Daskalakis
Fall 2026



Nash's Theorem



[Nash 1950]: Every finite game (i.e. game with a finite number of players and a finite number of pure strategies per player) has a Nash equilibrium.

- **Proof (previous lecture):** using Brouwer's fixed point theorem.
- **[Brouwer 1911]:** Every continuous function $f: D \rightarrow D$ from a convex compact set D to itself has a fixed point, i.e. some $x^* = f(x^*)$.

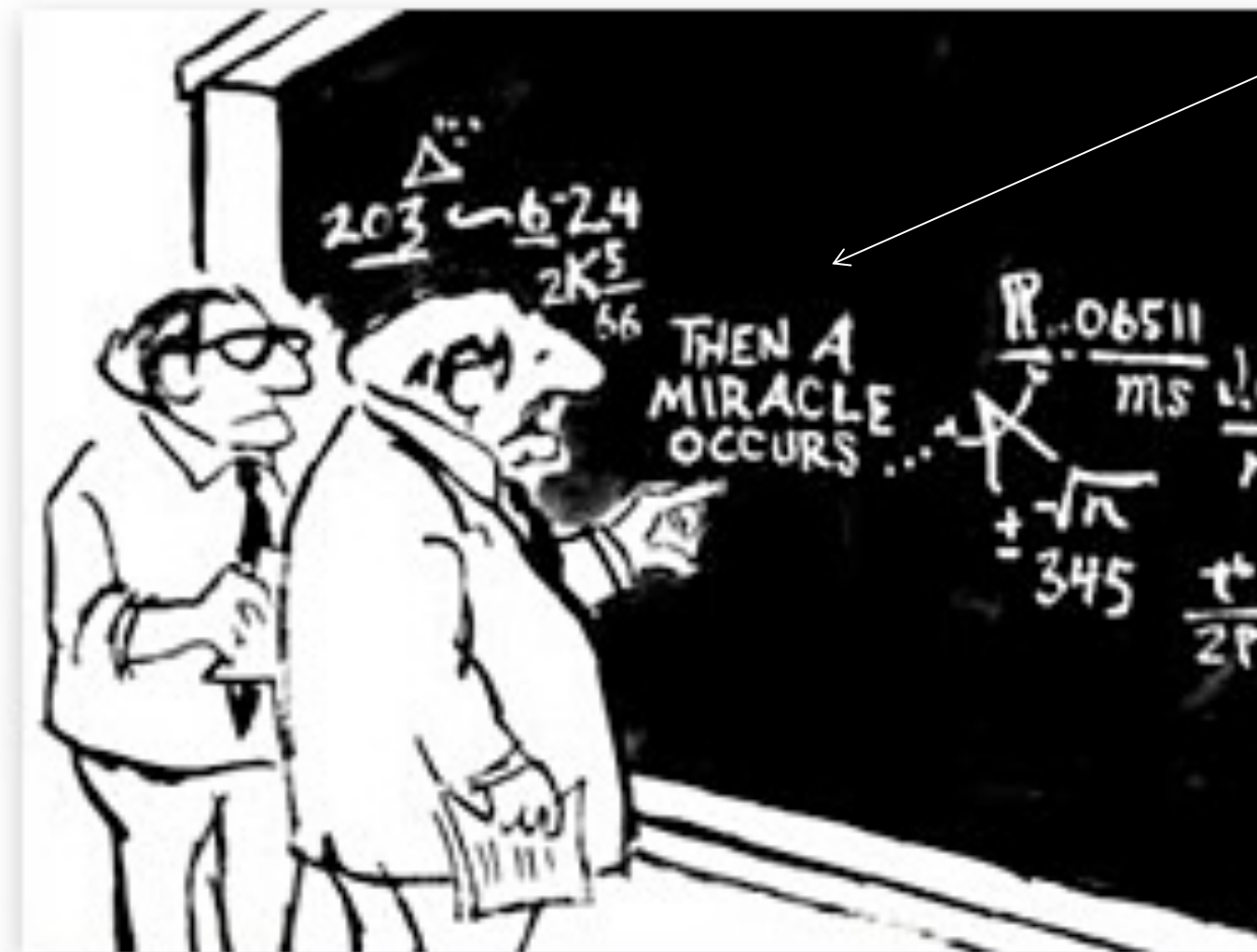


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- No simpler proof has been discovered for Nash in 75 years, and no polynomial-time algorithm has been discovered for either Nash equilibrium or Brouwer fixed point computation

Today's focus: understanding the non-constructive step



What step of Nash's proof and/or Brouwer's existence proof is not easy to simulate in polynomial-time?

Menu

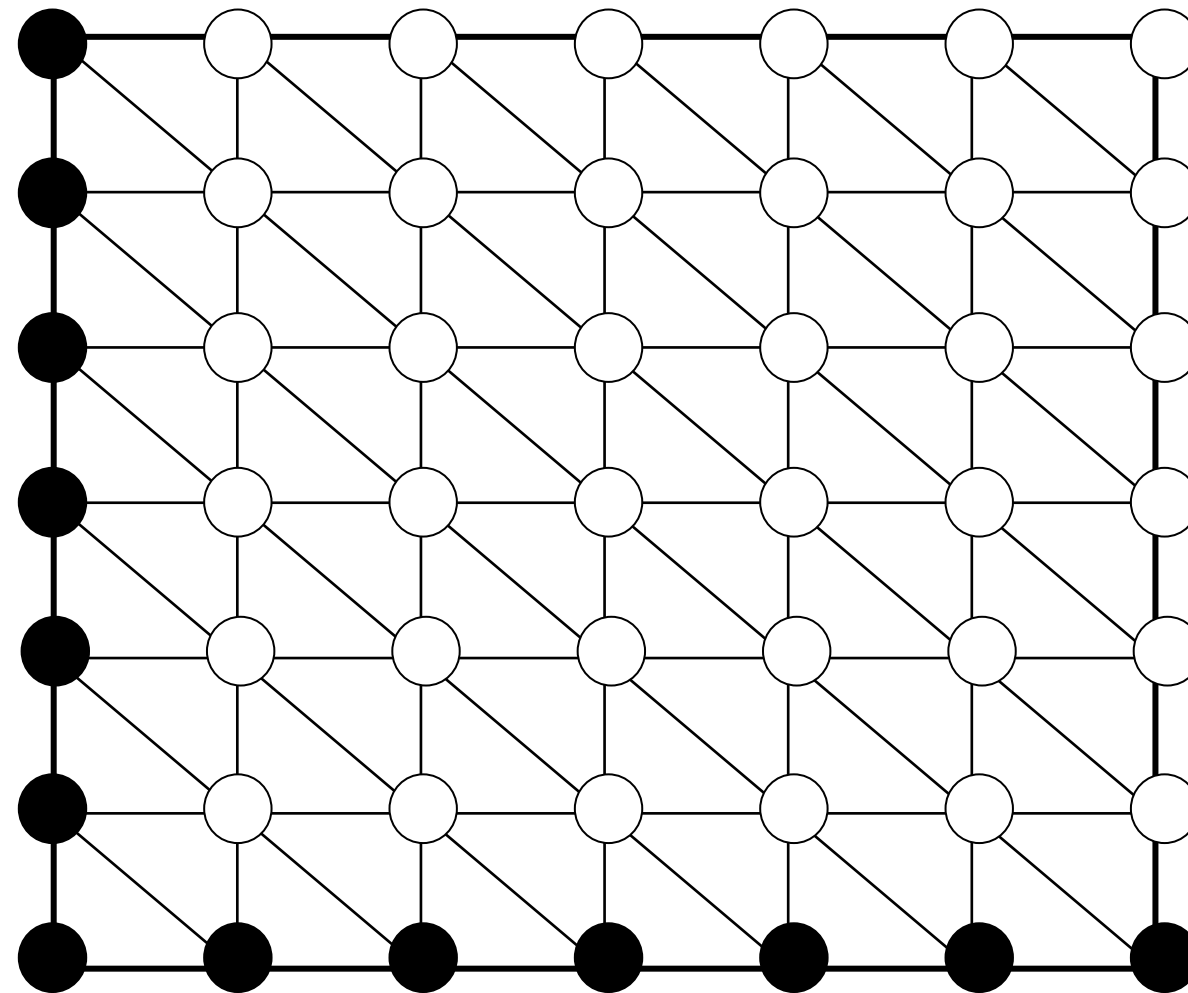
- **Refresher: Nash & Brouwer**
- Sperner's Lemma
- Brouwer via Sperner
- Sperner's Proof

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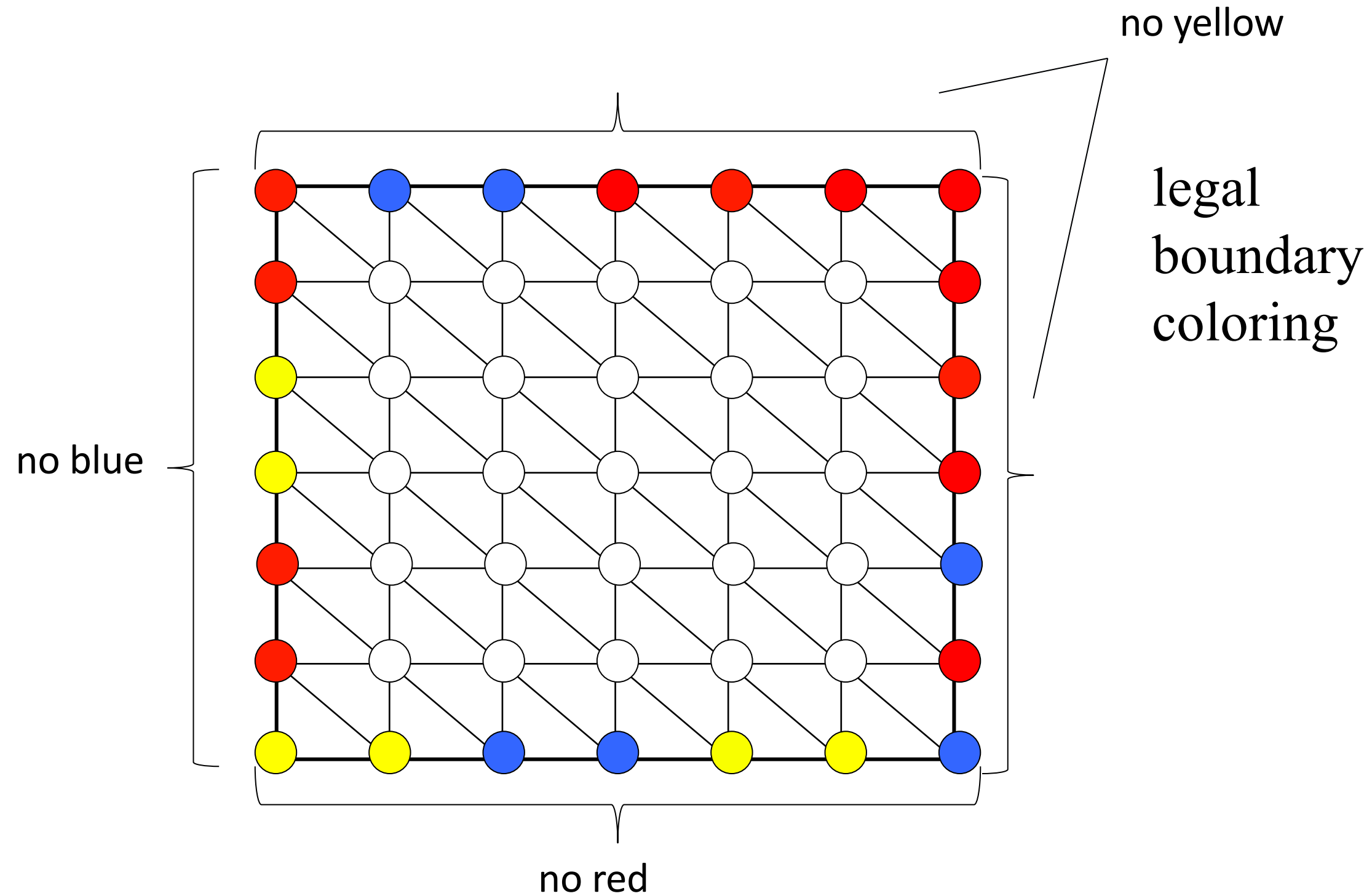
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Sperner's Lemma (2-d)

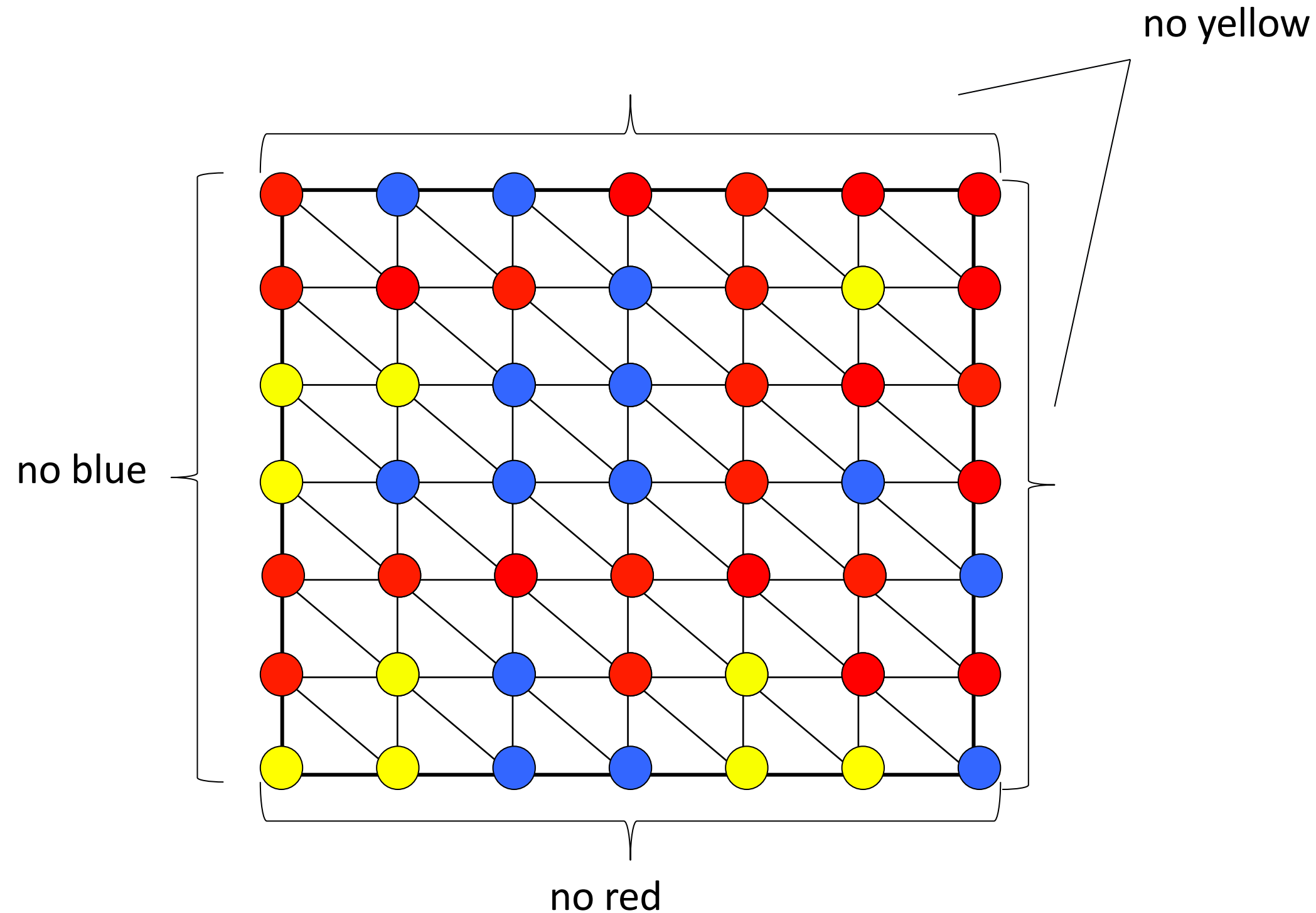


Sperner's Lemma (2-d)



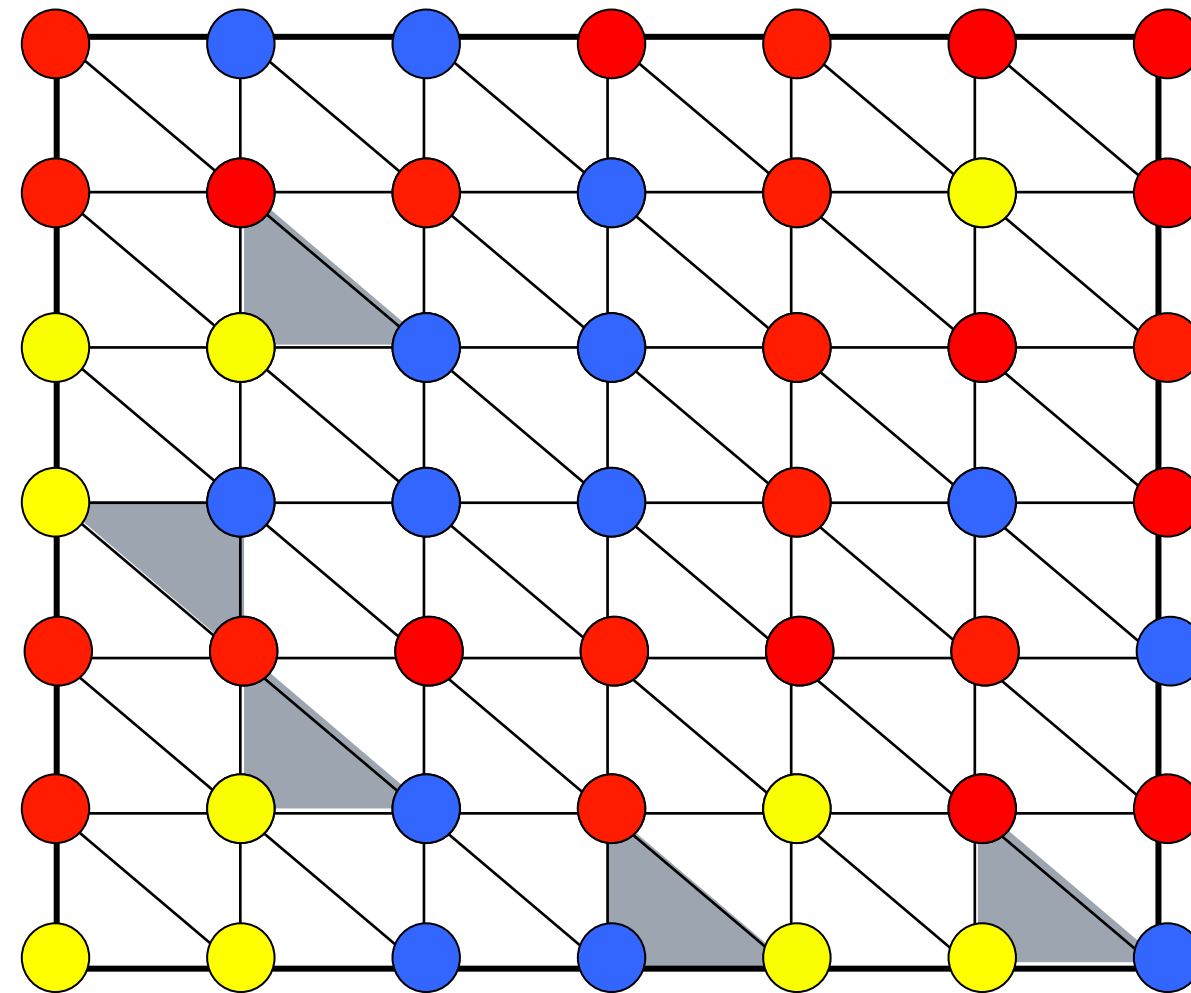
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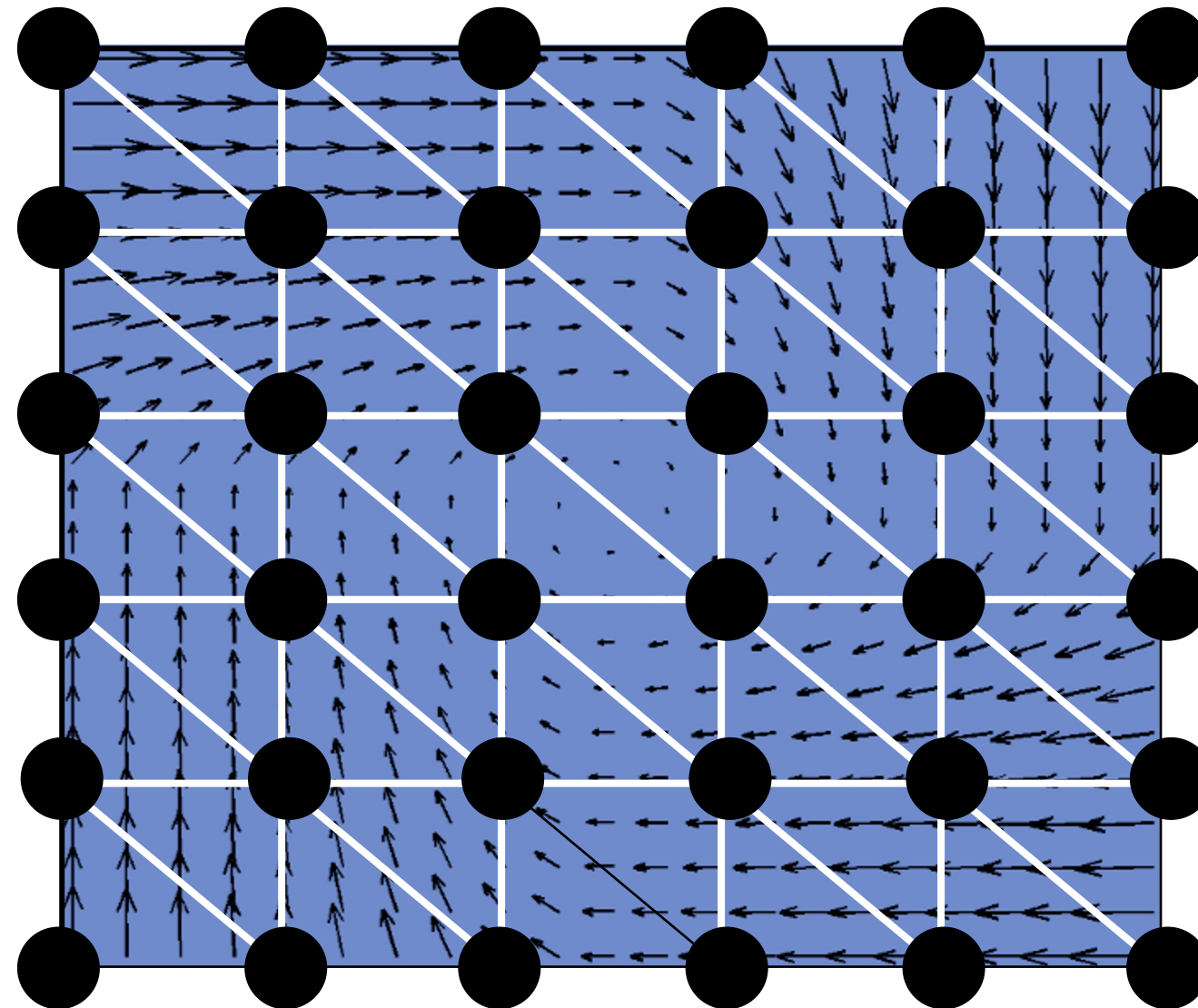
Sperner $\Rightarrow$ Brouwer

Sperner $\Rightarrow$ Brouwer (High-Level)

Given continuous $f : [0,1]^2 \rightarrow [0,1]^2$

1. For all ε , existence of approximate fixed point $|f(x)-x| < \varepsilon$, can be shown via Sperner's lemma.
2. Then use compactness.

For 1: Triangulate $[0,1]^2$;

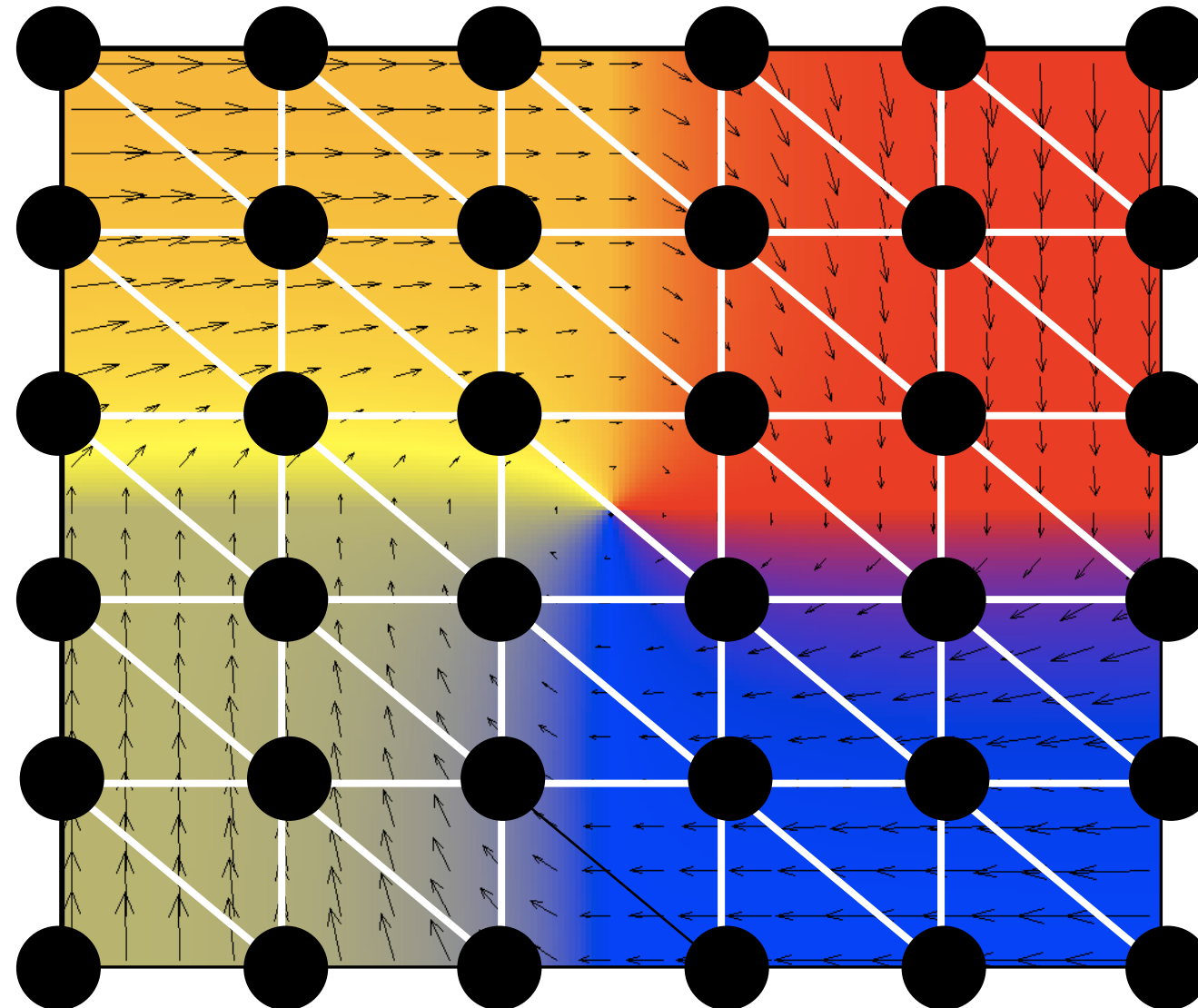
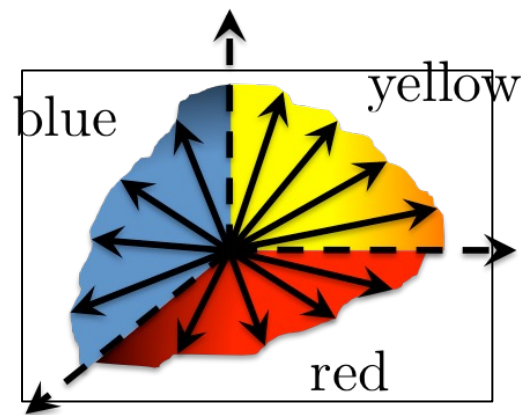


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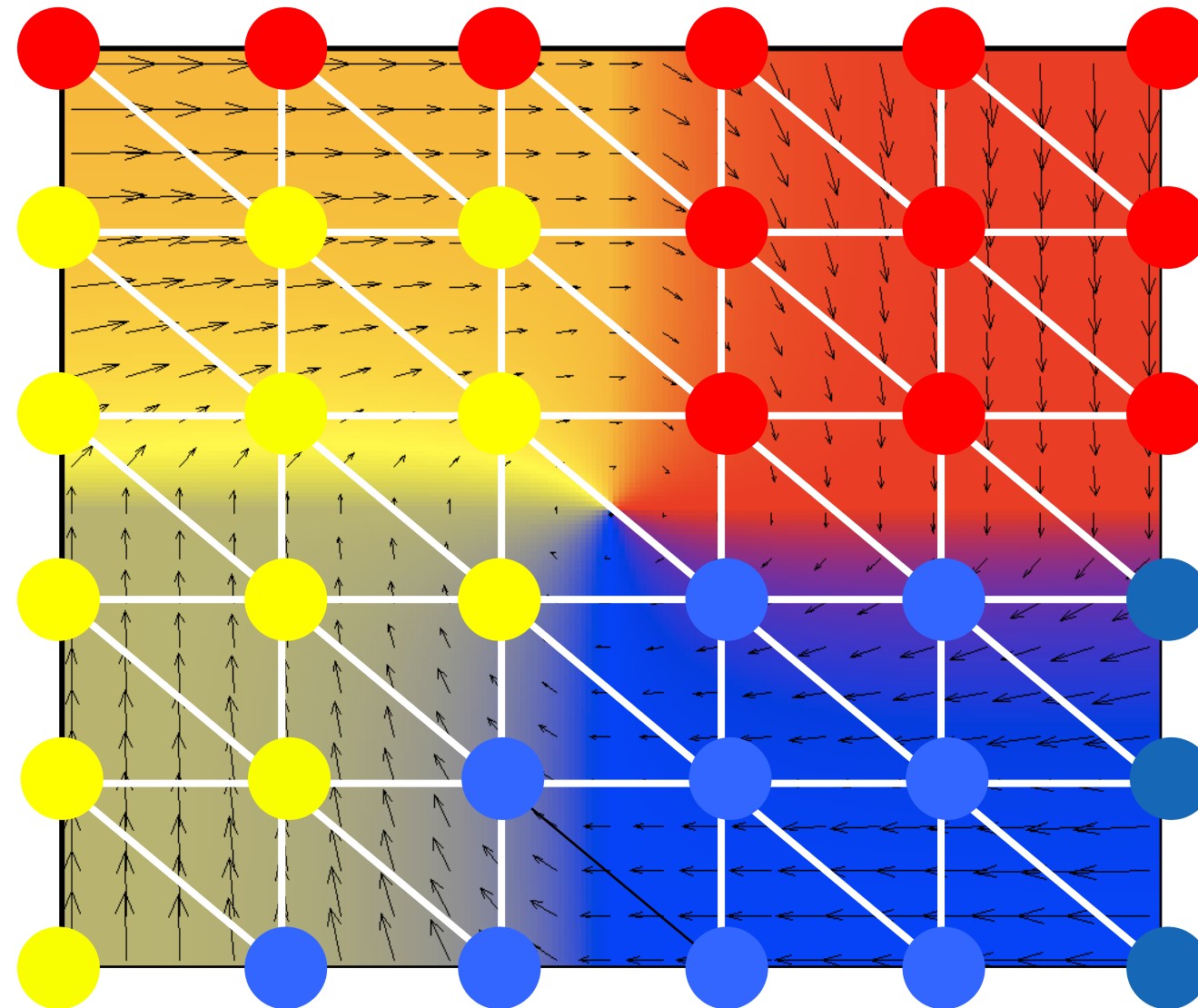
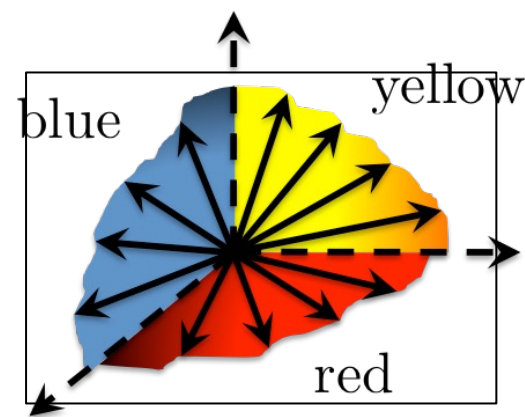


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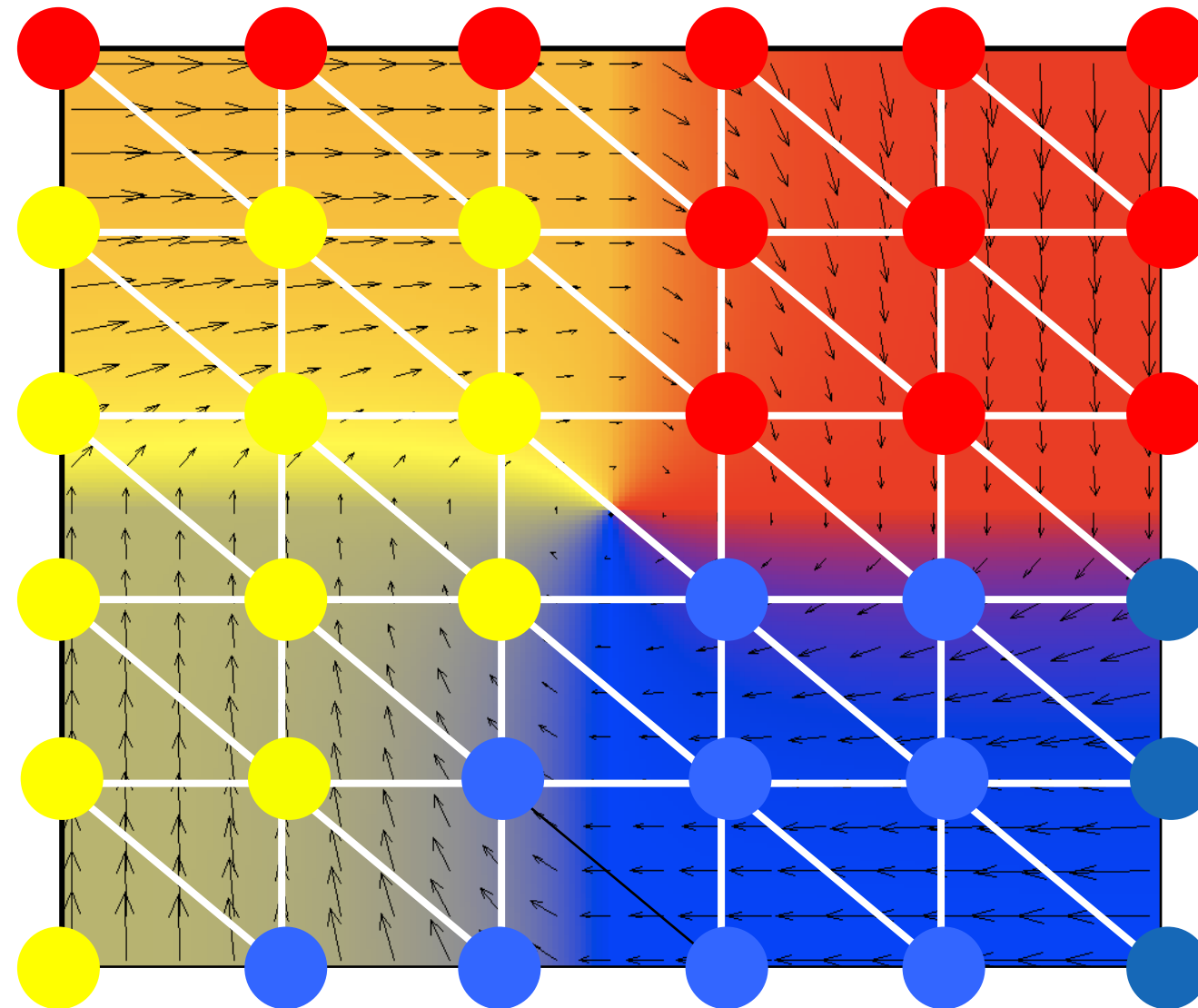
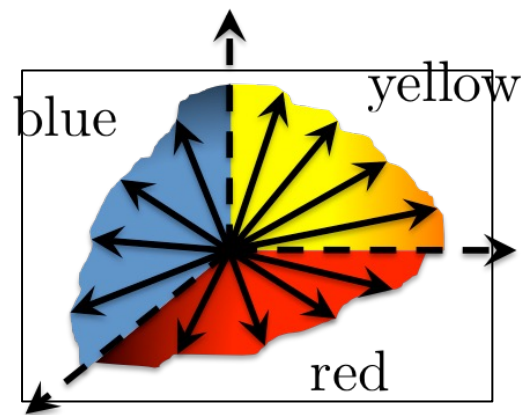


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1. For all ε , existence of approximate fixed point $|f(x)-x| < \varepsilon$, can be shown via Sperner's lemma.
2. Then use compactness.

For 1: Triangulate $[0,1]^2$;
then color points according
to the direction of $f(x)-x$;
apply Sperner and argue
trichromatic triangle
contains approximate
fixed points



2D-Brouwer on the Square

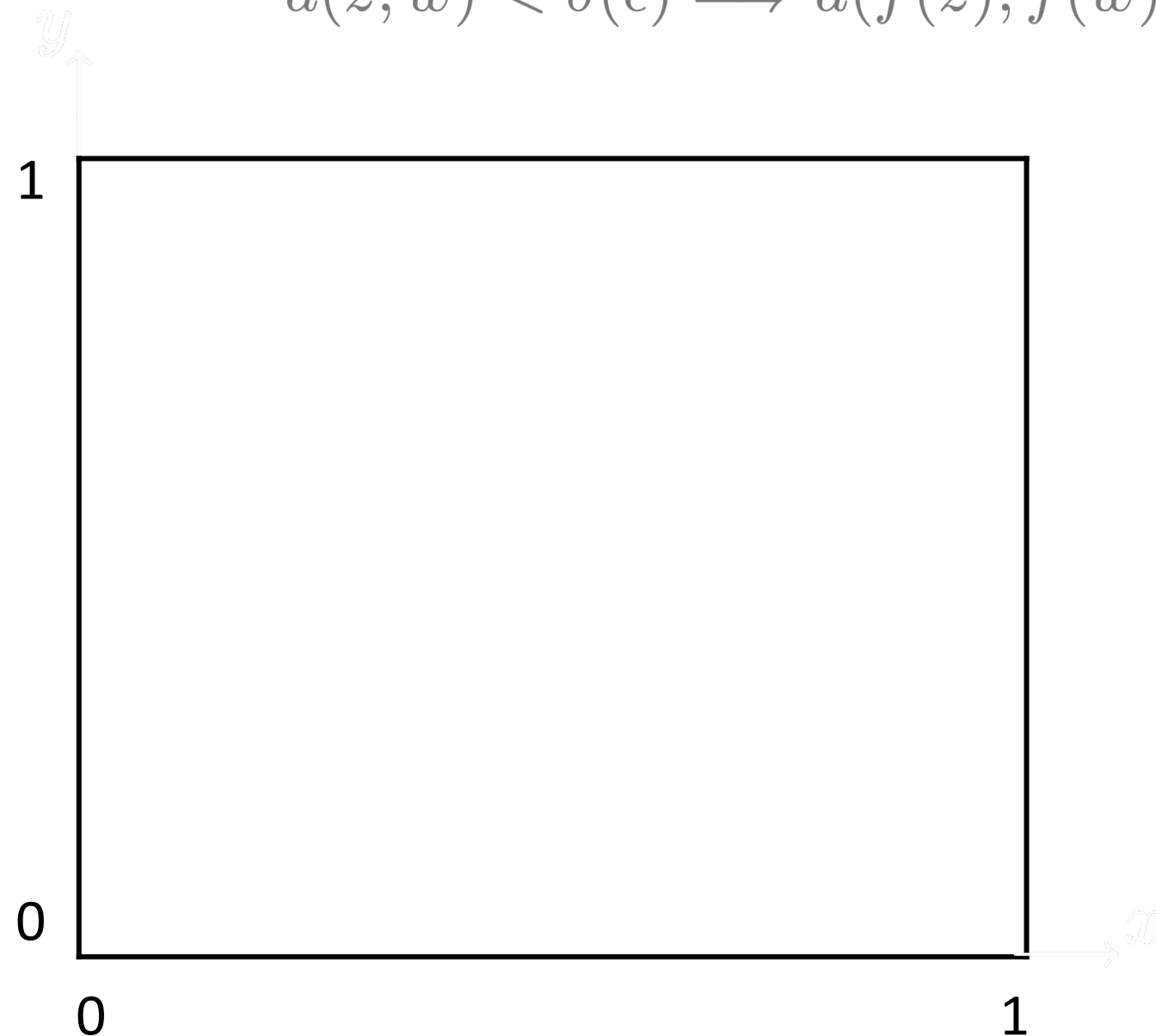
say d is the ℓ_∞ norm

Suppose $f: [0,1]^2 \rightarrow [0,1]^2$, continuous

└ must be uniformly continuous (by the Heine-Cantor theorem)

$$\forall \epsilon > 0, \exists \delta(\epsilon) > 0, s.t.$$

$$d(z, w) < \delta(\epsilon) \implies d(f(z), f(w)) < \epsilon$$



2D-Brouwer on the Square

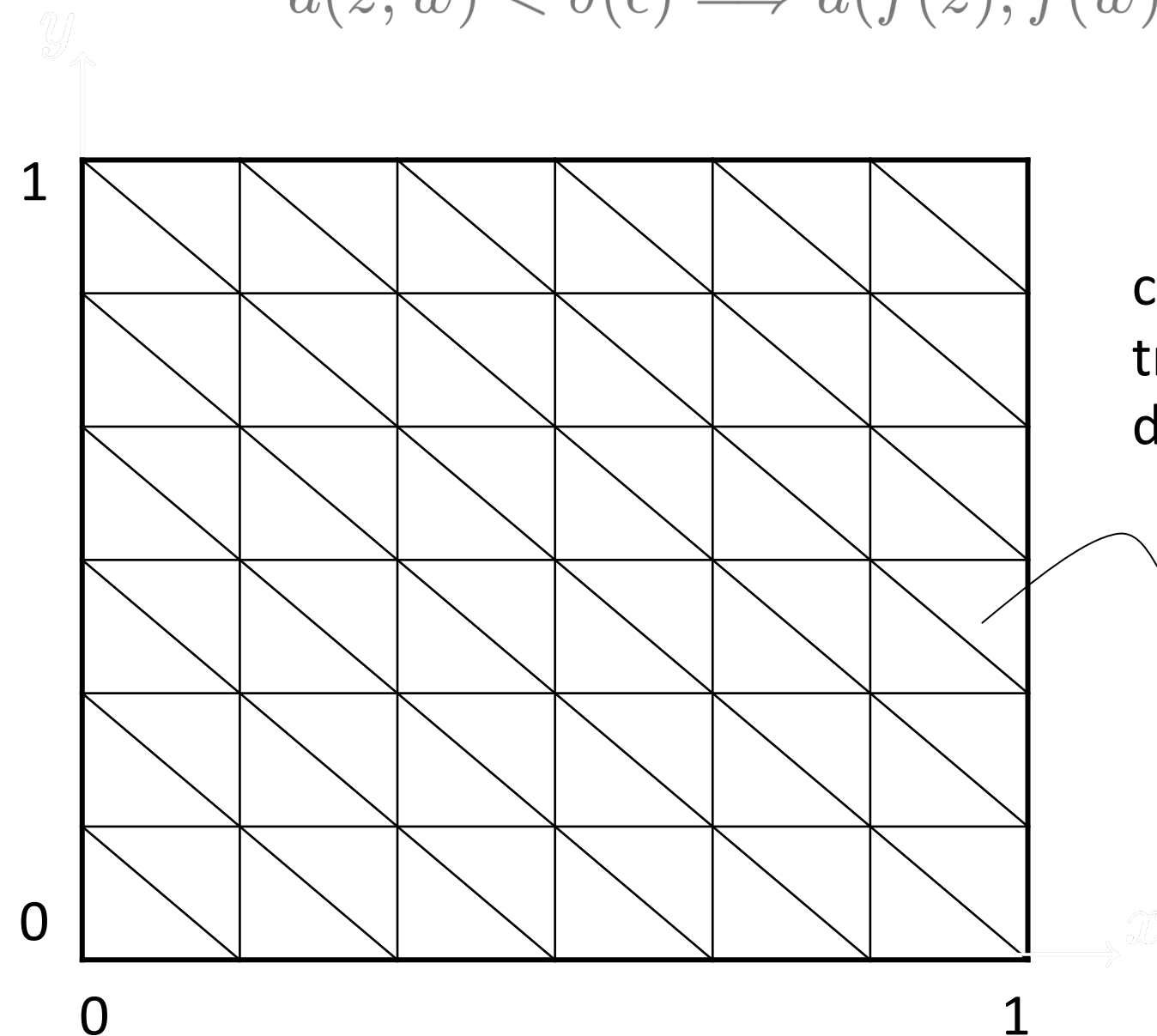
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choose some ϵ and
triangulate so that the
diameter of cells is

$$\delta < \delta(\epsilon)$$

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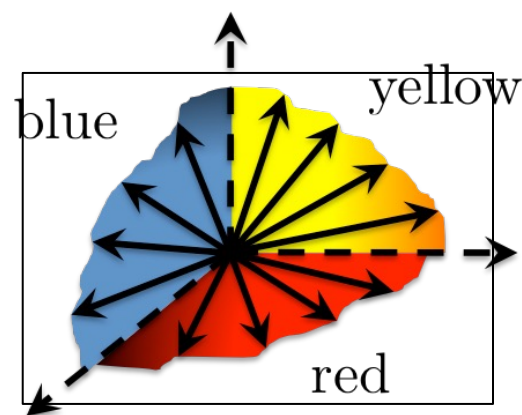
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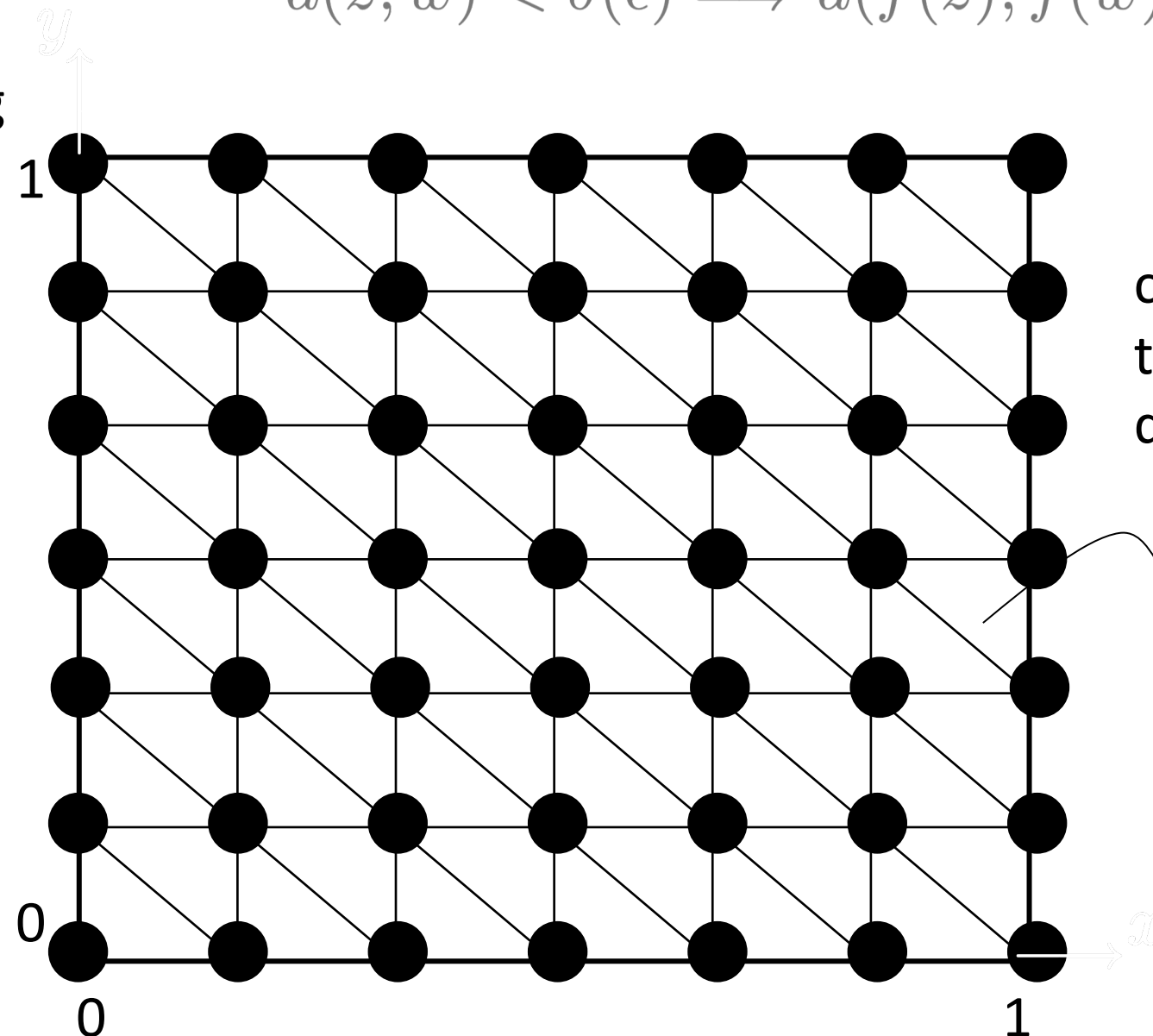
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color the nodes of the
triangulation according
to the direction of



$$f(x) - x$$



choose some ϵ and
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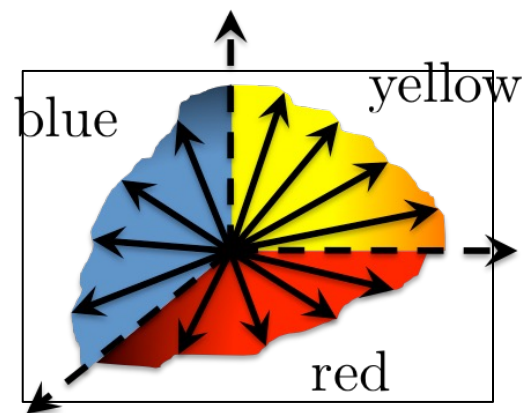
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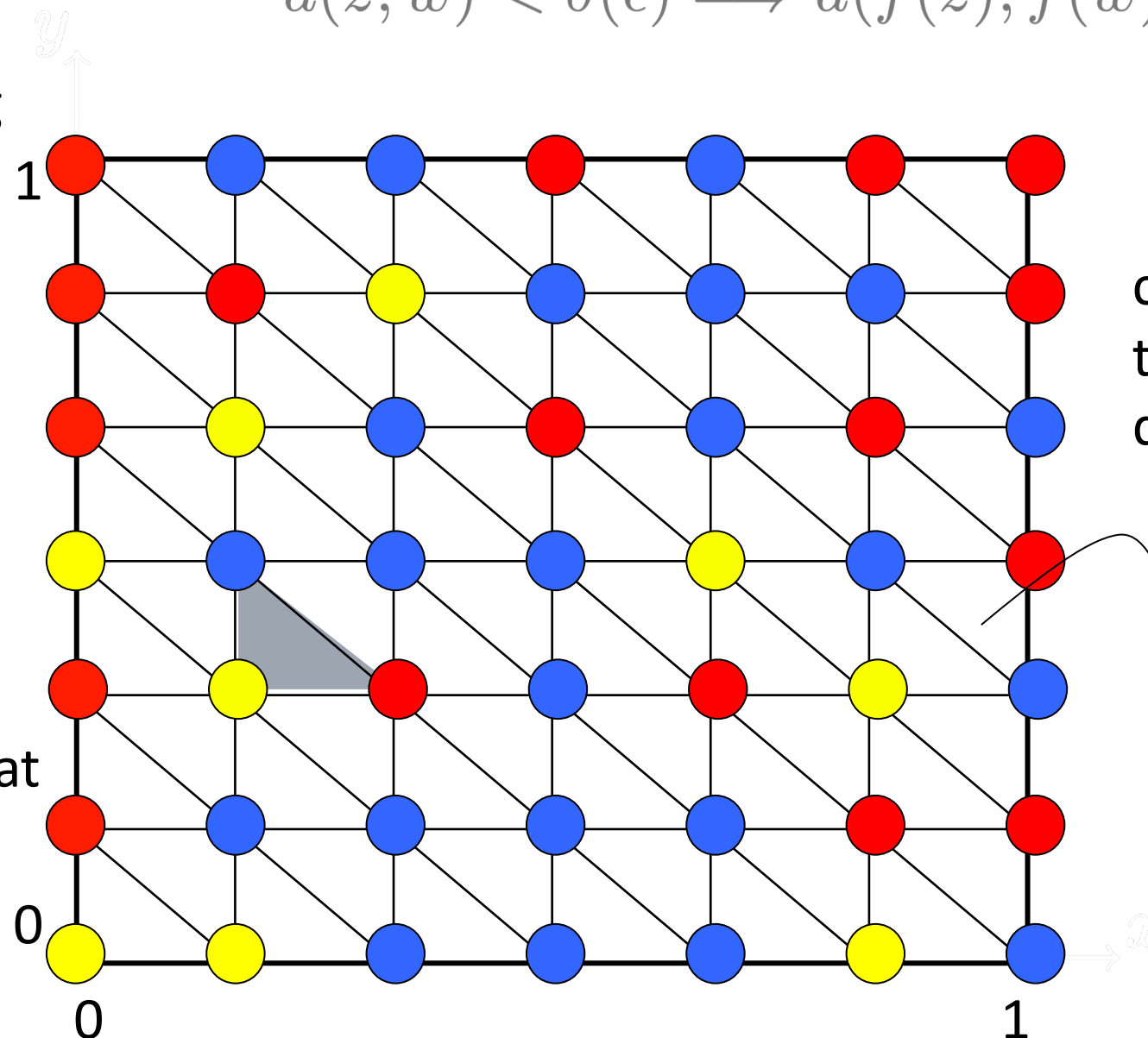
$$\forall \epsilon > 0, \exists \delta(\epsilon) > 0, s.t.$$

$$d(z, w) < \delta(\epsilon) \implies d(f(z), f(w)) < \epsilon$$

color the nodes of the triangulation according to the direction of



(tie-break at the boundary angles, so that the resulting coloring respects the boundary conditions required by Sperner's lemma)



choose some ϵ and triangulate so that the diameter of cells is

$$\delta < \delta(\epsilon)$$

find a trichromatic triangle, guaranteed by Sperner

2D-Brouwer on the Square

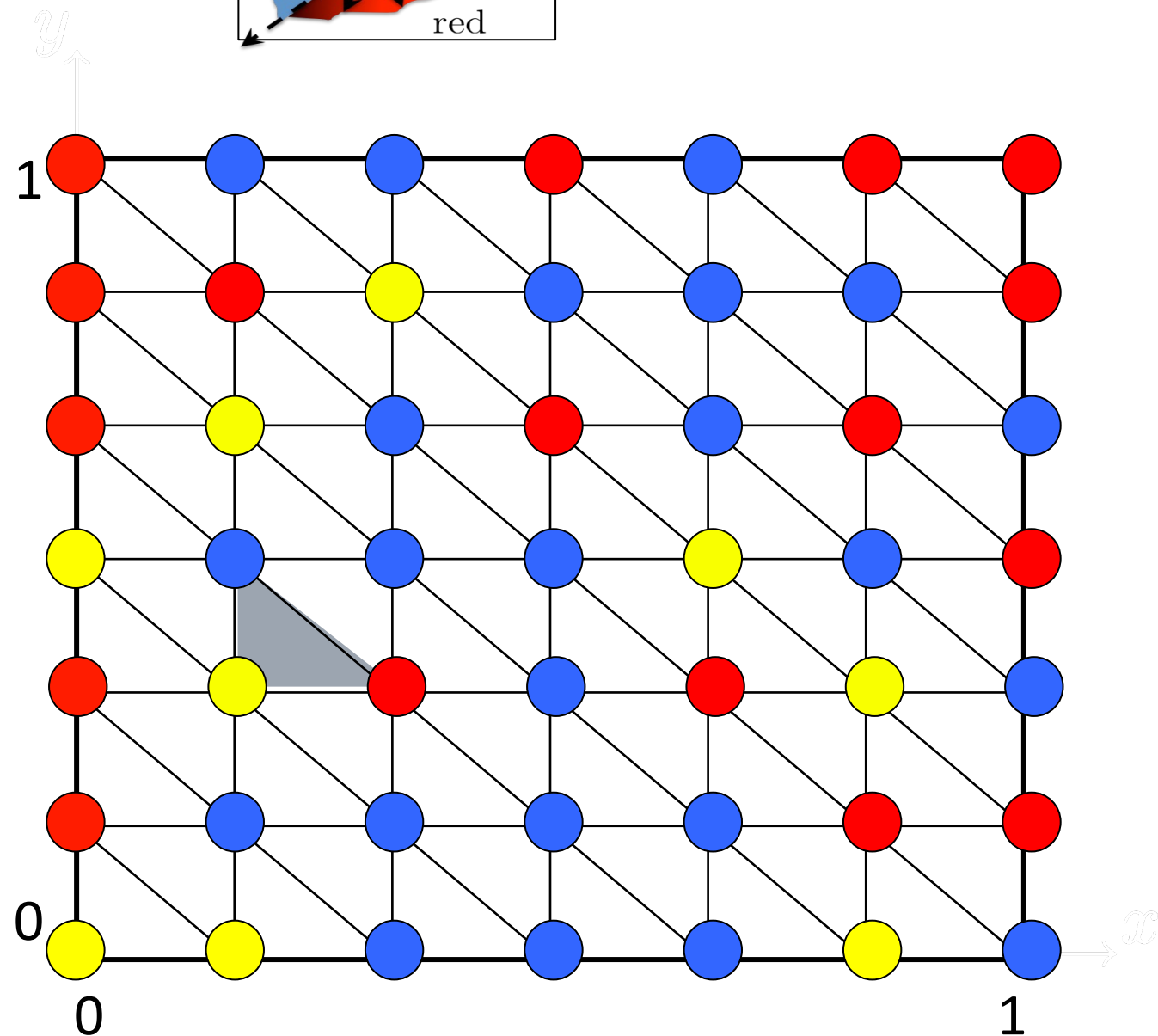
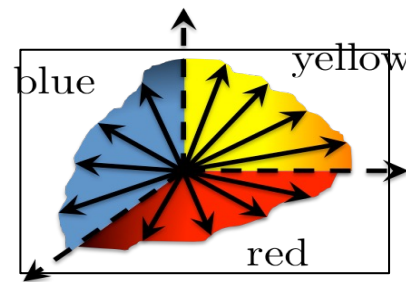
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Claim: If z^Y is the yellow corner of a trichromatic triangle, then

$$|f(z^Y) - z^Y|_\infty < \epsilon + \delta.$$

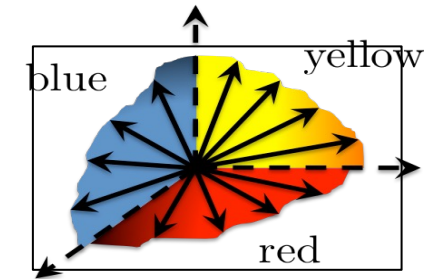
Proof of Claim

Claim: If z^Y is the yellow corner of a trichromatic triangle, then $|f(z^Y) - z^Y|_\infty < \epsilon + \delta$.

Proof: Let z^Y, z^R, z^B be the yellow/red/blue corners of a trichromatic triangle.

By the definition of the coloring, observe that the product of

$$(f(z^Y) - z^Y)_x \text{ and } (f(z^B) - z^B)_x \text{ is } \leq 0.$$

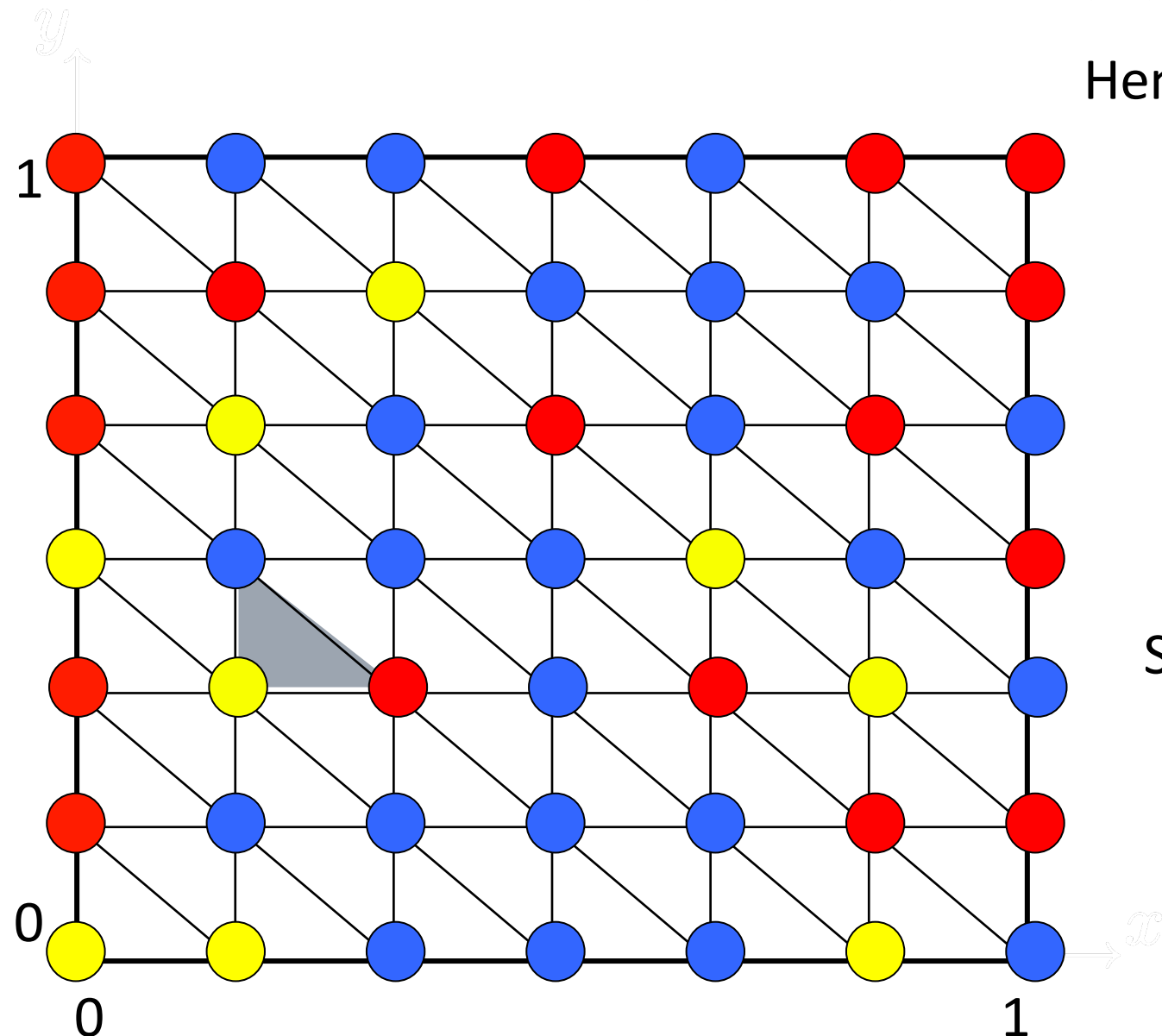


Hence:

$$\begin{aligned} |(f(z^Y) - z^Y)_x| &\leq |(f(z^Y) - z^Y)_x - (f(z^B) - z^B)_x| \\ &\leq |(f(z^Y) - f(z^B))_x| + |(z^Y - z^B)_x| \\ &\leq d(f(z^Y), f(z^B)) + d(z^Y, z^B) \\ &\leq \epsilon + \delta. \end{aligned}$$

Similarly, we can show:

$$|(f(z^Y) - z^Y)_y| \leq \epsilon + \delta.$$



2D-Brouwer on the Square

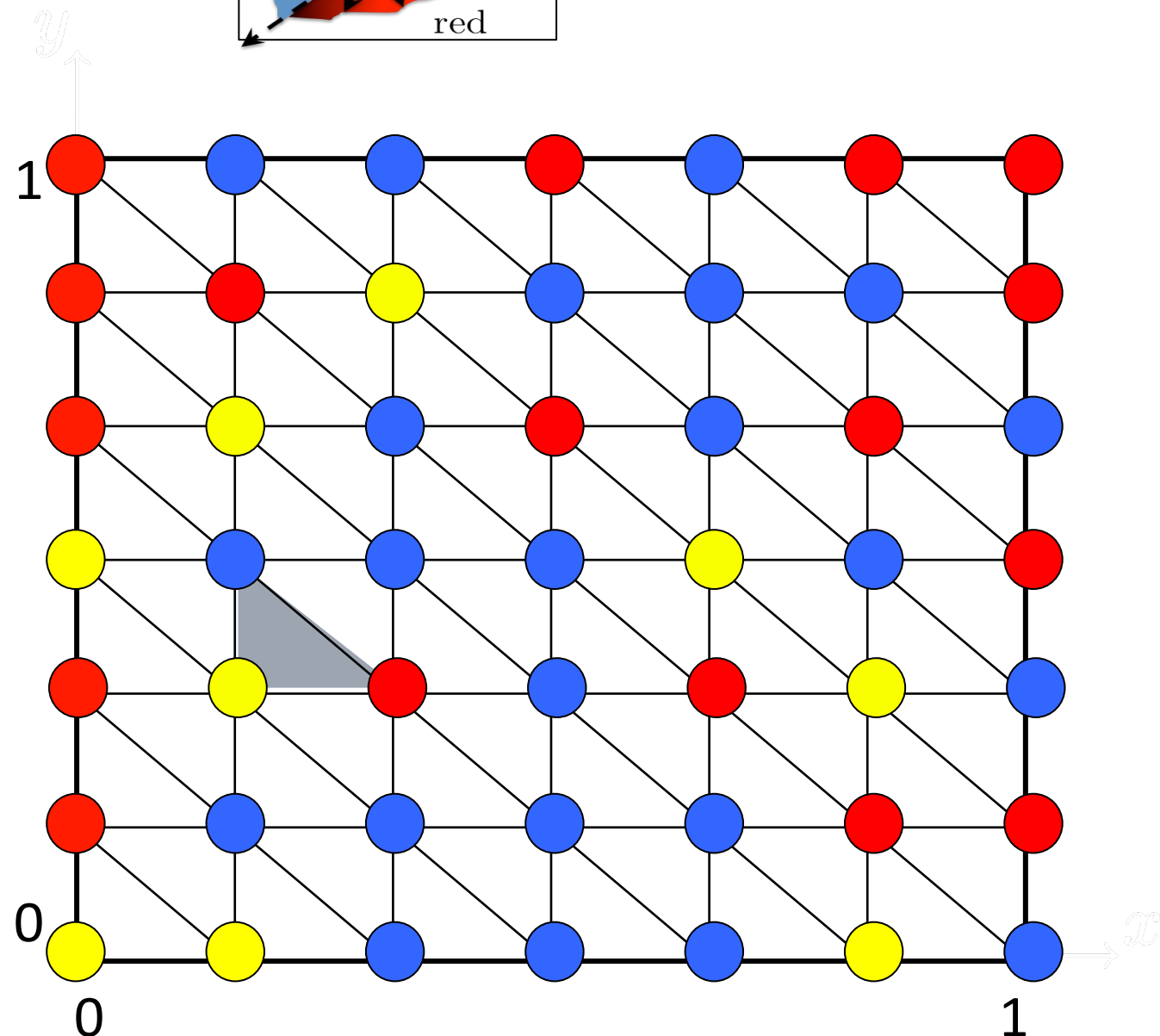
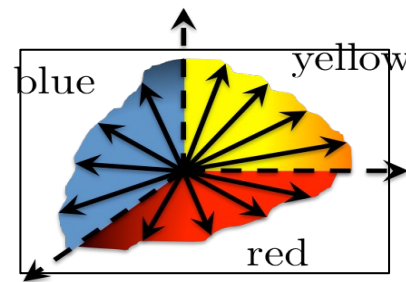
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Claim: If z^Y is the yellow corner of a trichromatic triangle, then

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Choosing $\delta = \min(\delta(\epsilon), \epsilon)$

$$|f(z^Y) - z^Y|_\infty < 2\epsilon.$$

2D-Brouwer on the Square

Finishing the proof of Brouwer's Theorem (Compactness):

- pick a sequence of epsilons: $\epsilon_i = 2^{-i}, i = 1, 2, \dots$
- define a sequence of triangulations of diameter: $\delta_i = \min(\delta(\epsilon_i), \epsilon_i), i = 1, 2, \dots$
- pick a trichromatic triangle in each triangulation, and call its yellow corner $z_i^Y, i = 1, 2, \dots$
- by compactness, this sequence has a converging subsequence $w_i, i = 1, 2, \dots$ with limit point w^*

Claim: $f(w^*) = w^*$.

Proof: Define the function $g(x) = d(f(x), x)$. Clearly, g is continuous since $d(\cdot, \cdot)$ is continuous and so is f . It follows from continuity that

$$g(w_i) \longrightarrow g(w^*), \text{ as } i \longrightarrow +\infty.$$

But $0 \leq g(w_i) \leq 2^{-i+1}$. Hence, $g(w_i) \longrightarrow 0$. It follows that $g(w^*) = 0$.

Therefore, $d(f(w^*), w^*) = 0 \implies f(w^*) = w^*$.



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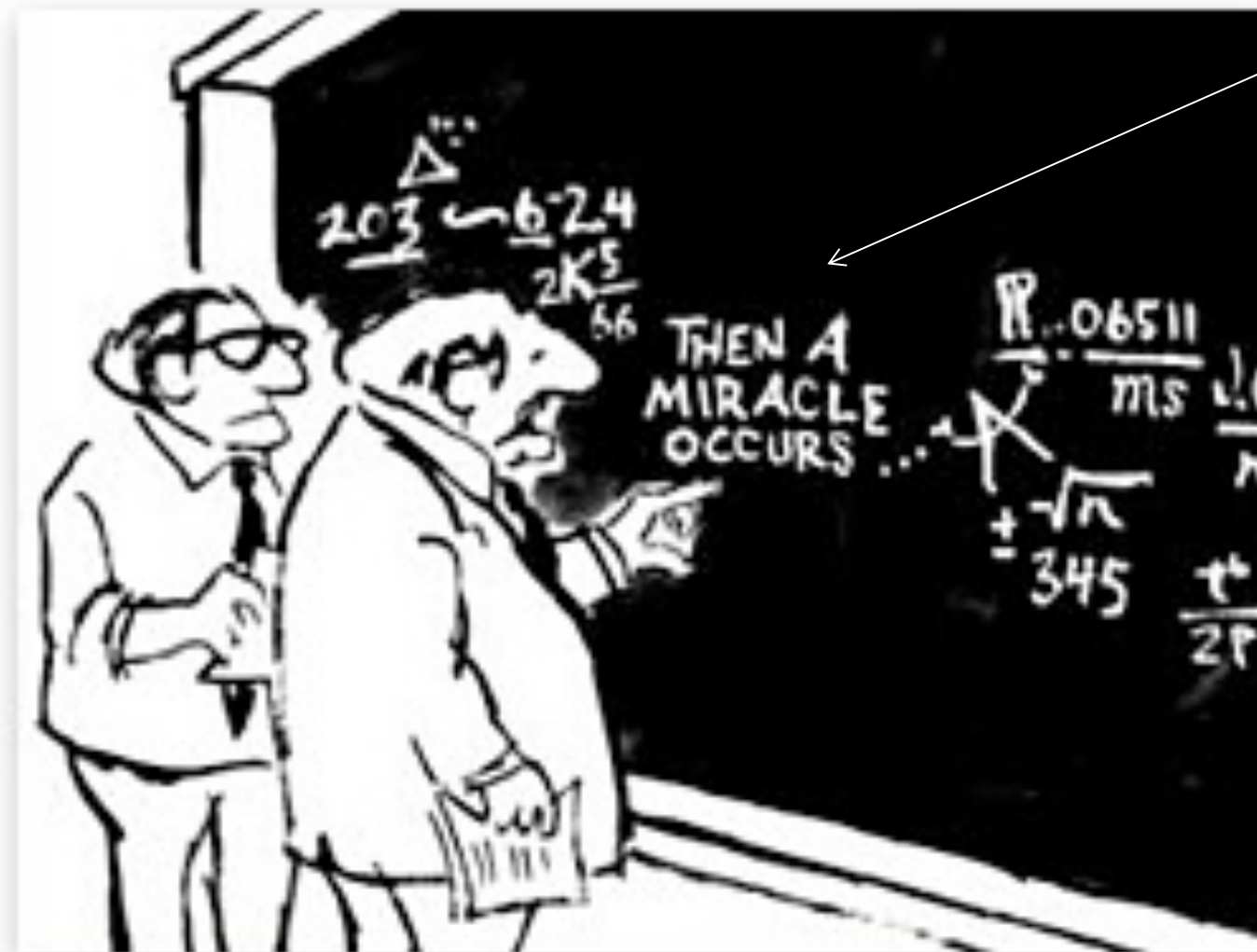
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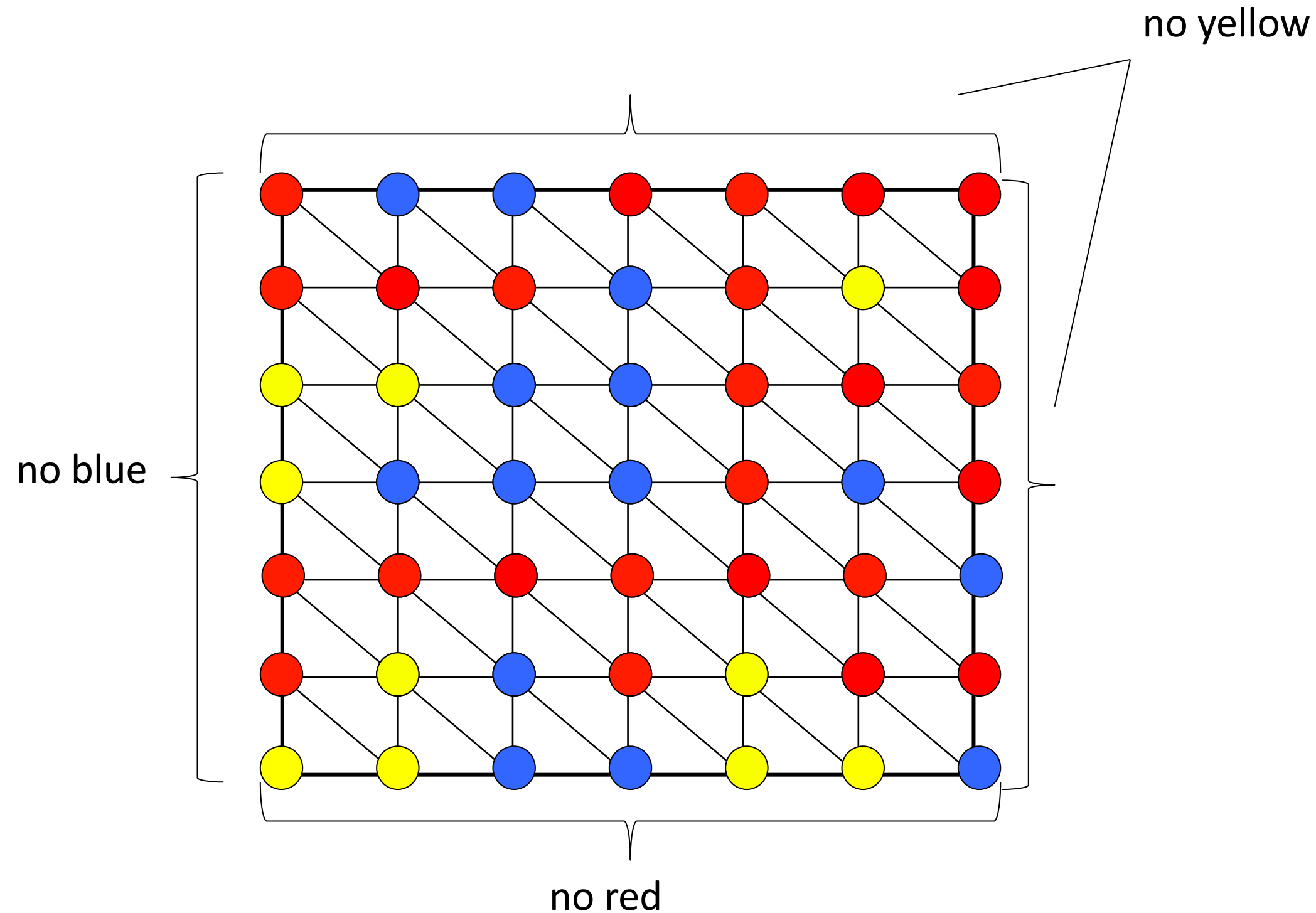
Today's focus: understanding the non-constructive step

What step of Nash's proof and/or Brouwer's existence proof is not easy to simulate in polynomial-time?



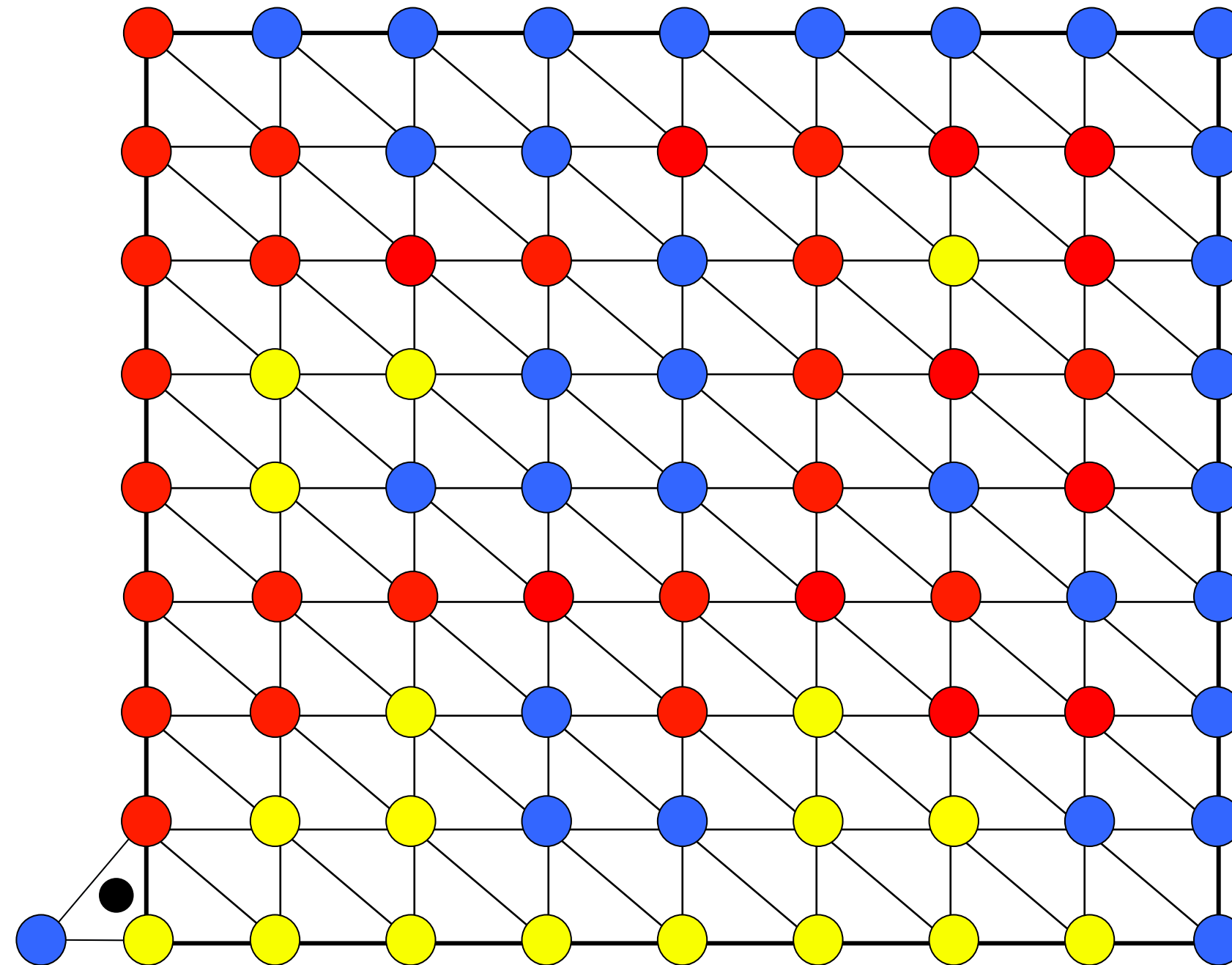
So far: Sperner's Theorem $\Rightarrow$ Brouwer's Theorem $\Rightarrow$ Nash's Theorem

Proof of Sperner's Lemma



[Sperner 1928]: Color the boundary using three colors in a legal way. No matter how the internal nodes are colored, there exists a tri-chromatic triangle. In fact an odd number of those.

Proof of Sperner's Lemma



For convenience we introduce an outer boundary, that does not create new tri-chromatic triangles.

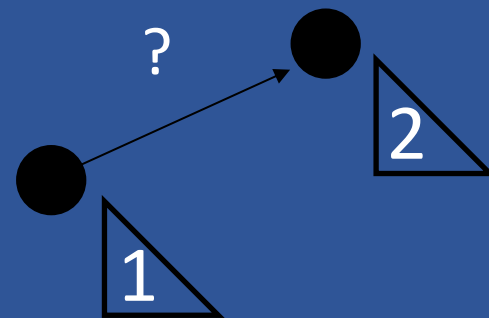
We also introduce an artificial tri-chromatic triangle.

Next we define a directed walk starting from the artificial tri-chromatic triangle.

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Proof of Sperner's Lemma

Transition Rule: If $\exists$ red - yellow door cross it with red on your left hand.



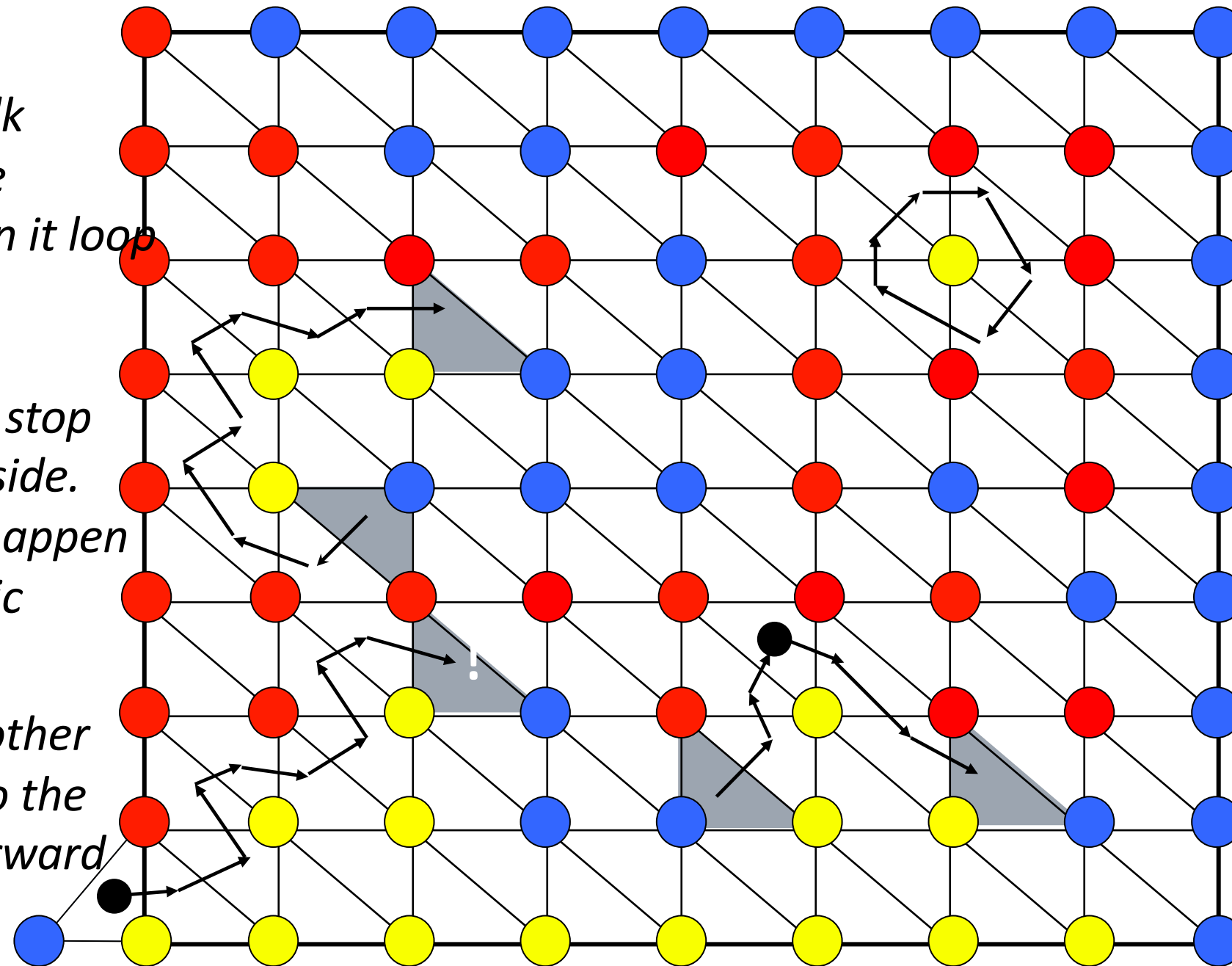
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Proof of Sperner's Lemma

Claim: *The walk cannot exit the square, nor can it loop into itself.*

Hence, it must stop somewhere inside. This can only happen at tri-chromatic triangle...

Starting from other triangles we do the same going forward or backward.



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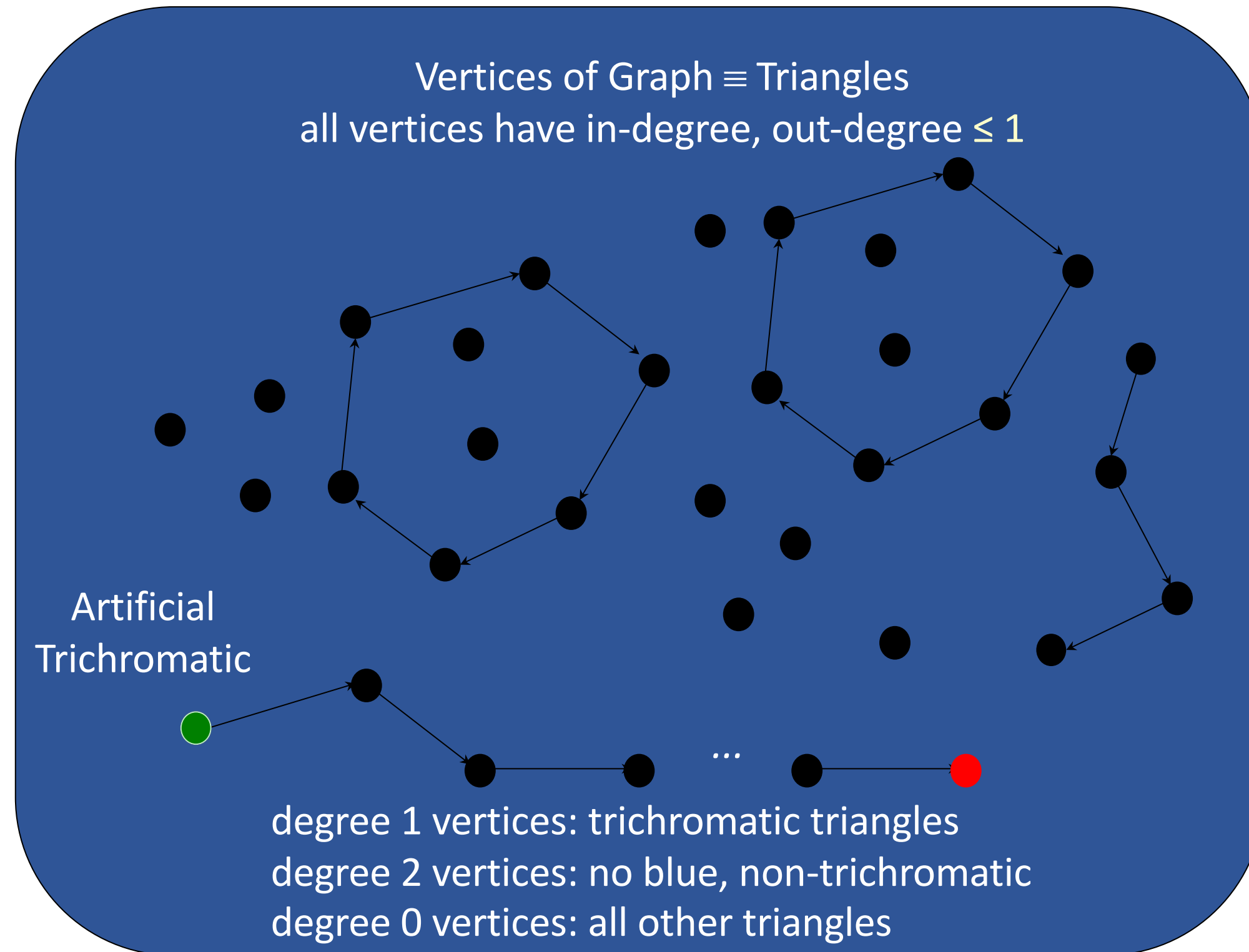
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Structure of Proof:

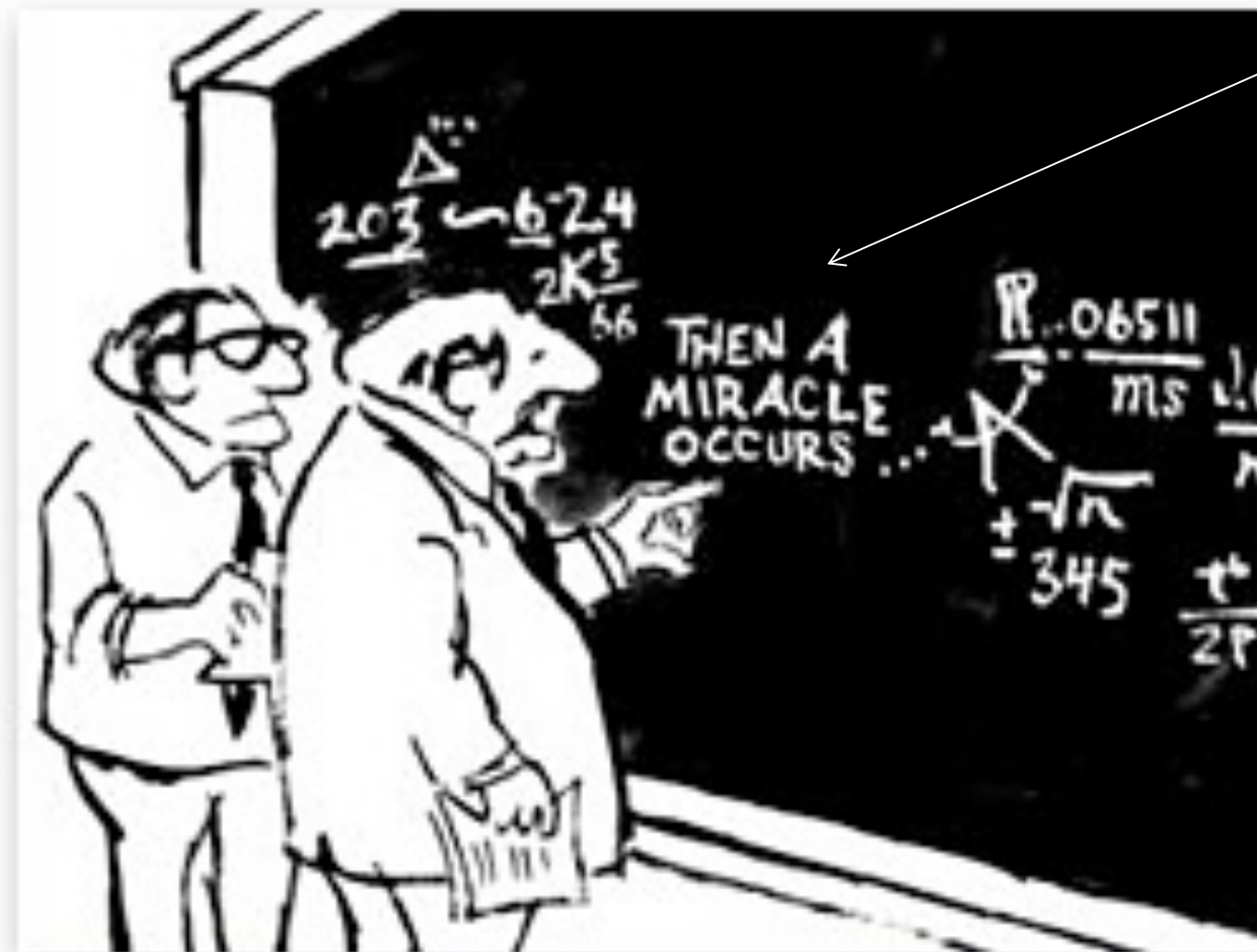
A directed parity argument



Proof: $\exists$ at least one trichromatic (artificial one) $\Rightarrow \exists$ another trichromatic
Also: degree 1 vertices are in pairs but one is fake $\Rightarrow \exists$ odd number of trichromatic!

So...what is the non-constructive step?

What step of Nash's proof is not easy to simulate in polynomial-time?

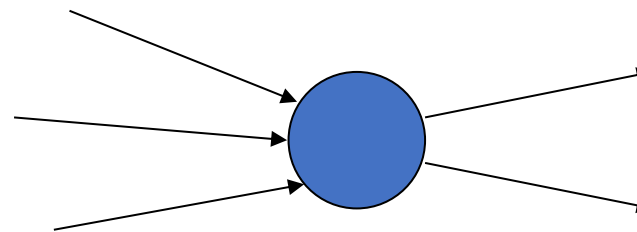


We have shown: Sperner's Theorem $\Rightarrow$ Brouwer's Theorem $\Rightarrow$ Nash's Theorem

The Non-Constructive Step

An easy parity lemma:

A directed graph with an unbalanced node (a node with indegree $\neq$ outdegree) must have another.



But, wait, why is this non-constructive?!?!

Given a directed graph and an unbalanced node, isn't it trivial to find another unbalanced node?

In some cases, the graph can be exponentially large in its description...

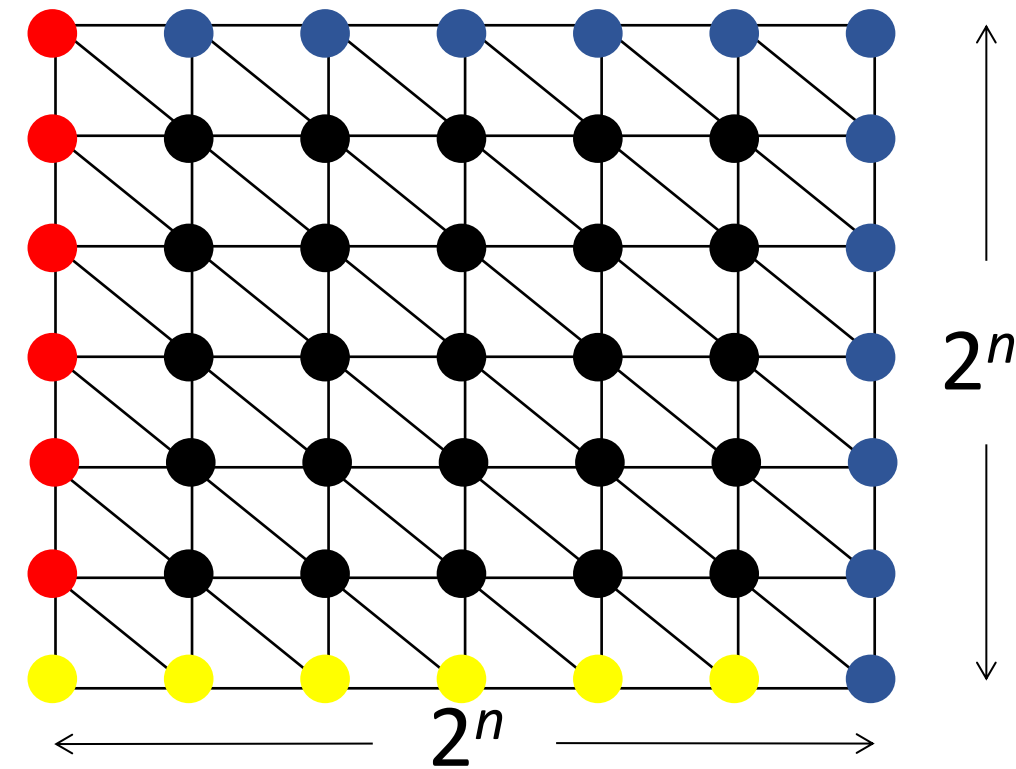
Example: next slide!

Computational Problem: SPERNER

INPUT:

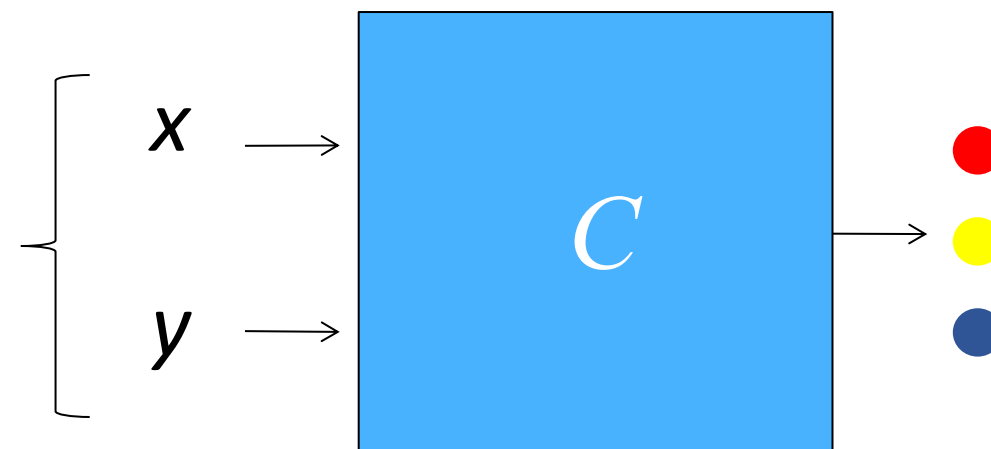
(i) n : specifies the size of a grid

(grid never written down!)



(ii) Imagine boundary has standard coloring shown above, while colors of internal vertices are given by a circuit:

input: the
coordinates
of a point
(n bits each)



OUTPUT: A tri-chromatic triangle

*exists because boundary coloring satisfies Sperner lemma constraints
but doing walk through grid to find one may take exponential time in n*